MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 8168 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode
See figure 1
Mounting position: Any
Weight: 27 ounces (840 grams) nominal

ABSOLUTE RATINGS:

Parameter:	F	Ef	Eb	Ec1	Ec2	Ib	Pg1	Pg2
Unit:	MHz	Vac	Vdc	Vdc	Vdc	mAdc	W	W
Maximum:								
AB1 or B1:	110	6.0±5%	3,000	-250	400	1,000	0	12
TEST CONDITIONS:	---	6.0	2,000	Adj	300	250	---	---

ABSOLUTE RATINGS:

Parameter:	Pp	T(seal)	T(anode core)	tk	Cooling
Unit:	W	°C	°C	sec(min)	Note 1
Maximum:					
AB1 or B1:	1,000	250	250	180	---
TEST CONDITIONS:	---	---	---	300	Note 2

GENERAL:

Qualification - Required

1/ Formerly 8168/4CX1000A

Ⓑ denotes changes

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MIL-STD-1311	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	8.1	9.9	Aac
1261	Electrode voltage (grid)	-		0.65	II	Ec1	-47	-60	Vdc
1256	Electrode current (screen)	-		0.65	II	Ic2	---	-10	mAdc
1266	Primary grid emission (control)	-	Ec1/Pg1 = 1 W; a = g2 = k = 0 Vdc; E(avg) reverse = 25 V; t = 60	0.65	II	Isg1	---	-10	µAdc
1266	Primary grid emission (screen)	-	Ec2/Pg2 = 12 W; a = g1 = k = 0 Vdc; E(avg) reverse = 50 V; t = 60	0.65	II	Isg2	---	-25	µAdc
---	Ion current	5	Eb = -67.5 Vdc; Ec2 = 300 Vdc; Ec1/Ic2 = 20 mAdc	0.65	II	Iz	---	10	µAdc
(B) 1231	Pulse emission (1) (sinusoid)	9	eb = ec2 = ec1 = 500v ; prr = 10 (min)	0.65	II	is	65	---	a
(B) 1231	Pulse emission (2)	9	eb = ec2 = ec1 = etd; etd/is = 65a; prr = 10 (min)	0.65	II	etd	---	500	v
1366	Interelement leakage resistance, cold	- 6	g1 to k; no other voltages g1 to g2; no other voltages	0.65	II	R R	50 50	--- ---	Meg Meg
	Quality conformance inspection, part 2								
1316	Amplification factor	-	g1 to g2; Ec2 = 300 Vdc; Ic2 = 40 mAdc; t = 30; anode floating	---	---	Mu	3.2	4.5	---
1331	Direct - interelectrode capacitance	-		---	---	Cgp Cin Cout	--- 75.0 10.8	0.022 88.0 12.6	pF pF pF
	Quality conformance inspection, part 3								
---	Life test	-	Group C; class AB amp; F = 1 to 32 MHz; Eb = 2,900 Vdc; Ec2 = 350 Vdc; Ec1/Ibo = 250 mAdc; Eg1/Ib = 800 mAdc; t = 500 hours	---	---	Po	1,200	---	W (useful)
1143	Forced cooling	3, 7	Ec1/Ib = 500 mAdc	---	---	T(anode core)	---	250	°C
1155	Coolant-pressure drop versus coolant flow	4, 7	No voltages	---	---	---	---	0.25	In.H_2O

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(see note 1)

METHOD	REQUIREMENT OR TEST	NOTE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
1032	Vibration, mechanical	7, 8	Ef = 6.0 V, ac or dc; Ebb = 2,500 Vdc; Ec2 = 400 Vdc; Rp = 1,000 ohms; Ec1/Ib = 250 mAdc; accel = 10 G peak (min) or DA = 0.25 inch (min) below 28 Hz; F = 10 to 500 to 10 Hz	---	---	Ep	---	30	Vac
(8) 2214	RF useful output power	7	Class AB1 amplifier; F = 1 MHz (min); Eb = 3,000 Vdc; Ec2 = 350 Vdc (max); Ec1/Ib = 250 mAdc with Pd = 0 W; Eg1/Ib = 1,000 mAdc (max); Ic1 = 0 mAdc; Ic2 = 30 mAdc (max); t = 15 minutes (min)	---	---	Po	1,500	---	W (useful)
1042	Shock, specified pulse	7	Test condition A	---	---	---	---	---	---
---	Vibration, mechanical and shock, specified pulse end points.								
1256	Electrode current (ion)	7		---	---	Iz	---	10	μAdc
1261	Electrode voltage (grid)	7		---	---	Ec1	-47	-60	Vdc

NOTES:

1. In all cases of operation, forced-air cooling shall be applied and the maximum seal and anode core temperature rating shall not be exceeded. An air-system socket and air director (chimney), such as the EIMAC SK-800 series sockets with SK-806 chimney, or equivalents, should be used. The cooling table is based on installation of the tube in the SK-800/SK-806 socket/chimney combination, with air at a maximum of 40°C. The static-pressure drop figures shown are for the tube/socket/chimney combination, with airflowing in a base-to-anode direction. At very high frequency, increased airflow will be required. Cooling airflow shall be maintained during standby periods when only the heater voltage is applied to the tube. Airflow shall be applied before or simultaneously with the electrode voltages, and may be removed with them. Where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial.

Sea level			10,000 Feet		
Anode diss.	Airflow (cfm)	Approx. press. drop ($In.H_2O$)	Anode diss.	Airflow (cfm)	Approx. press. drop ($In.H_2O$)
600 W	13	0.05	600 W	19	0.08
800	17.5	0.10	800	26	0.15
1,000	25	0.20	1,000	37	0.30

2. In all electrical tests involving application of heater voltage, the use of the air-system socket/chimney combination SK-800/SK-806 series, or equivalents, shall be allowable. Forced-air cooling is permitted at the rate of 25 cfm maximum for the base and anode of sea level, rising to a maximum of 37 cfm at 10,000 feet elevation. A separate source may be used for the base and anode but neither shall exceed the totals specified above. Cooling air shall not have a temperature less than 20°C.

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NOTES: - Continued

3. The cooling test shall be made as follows: An SK-800 series socket shall be mounted in an infinite baffle system, or equivalent. The tube under test shall be mounted in the socket, with an SK-806 chimney mounted so that all air passing in a base-to-anode direction shall pass through the anode cooler of the tube. At an ambient temperature of not less than 20°C nor more than 40°C, both base and anode shall be cooled by applying an airflow of 25 cfm maximum at sea level. At the specified test conditions, the anode core temperature shall not exceed the specified limit. Temperature shall be measured by means of a thermocouple, located as follows: The termocouple shall be embedded in the top of the anode core, adjacent to the cooler, by means of drilling a small hole, shallow enough so that the tube vacuum shall not be lost, placing the welded thermocouple junction therein, and then peening the edges of the hole so as to hold the thermocouple firmly in place. Good electrical continuity between the thermocouple and the metal of the anode core shall be demonstrated before the cooling test is performed.

4. An infinite baffle system, or equivalent, with an airflow of 25 cfm at sea level shall be used. The static-pressure drop is measured across the tube/socket/chimney combination.

5. This test shall be the first test performed after the holding period. The tube shall be connected as an ion gauge with the specified potentials applied, and the highest initial reading shall be taken as the test value. Ion current is read with a microammeter in series with the negative anode lead. Except for Ef, all voltages shall be applied simultaneously, with an automatic current regulator controlling Ic2 to the specified value.

6. Test to be made with a potential of 50 ± 5 Vdc applied between the test points. An instrument such as the General Radio Megometer (Model 1862B), or equivalent, may be used.

(B) 7. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

8. Each tube under test shall be subjected to one sweep cycle in each of the three axes X, Y, and Z. One sweep cycle (10 to 500 and back to 10 Hz) shall be covered in 6 to 12 minutes. The specified voltages shall be applied during the test, using the basic test circuit shown, see figure 2. Tubes found to electrically oscillate for causes other than vibration shall not be tested nor rejected on this test. Each tube shall be vibrated for 60 seconds at the frequency which gives the maximum vibration (noise) output voltage in each of the three axes. If at the end of the 60 seconds the vibration output voltage is increasing, the vibration shall be continued until there is no further increase. The tubes shall not show noise voltage output in excess of the maximum limit specified, except one intermittent short per tube shall be allowable during this test. In addition to reading noise voltage on the specified VTVM, or equivalent, a permanent recording shall be made using a good quality recorder to produce a plot of noise voltage versus frequency. Noise voltage amplifiers used with the recorder shall have a ± 1 dB frequency response over the range to be measured and the overall recording equipment shall be capable of fast response in order to show sharp noise voltage spikes resulting from internal tube resonances or other phenomenon. Prominent noise peaks indicated on the recording shall be individually investigated by fixed-frequency operation, and the 60-second operation shall be made at the frequency of highest noise as so selected.

9. Method 1231, pulse emission (1) or pulse emission (2) tests may be performed alternately as specified herein. Only one of these tests shall be performed.

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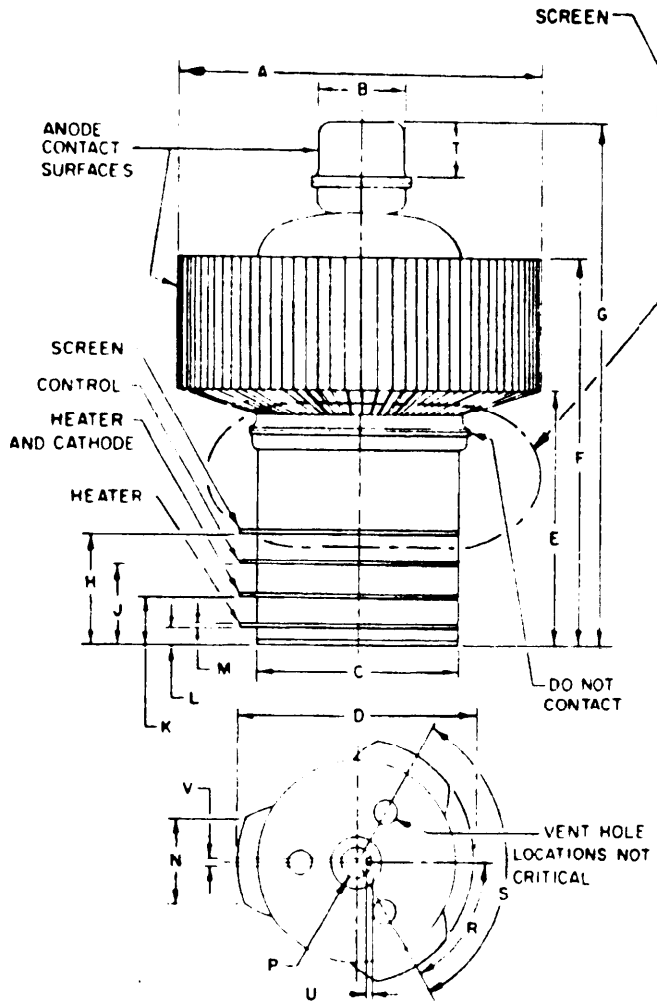
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
C	1.870 (47.50)	1.900 (48.26)
G	4.600(116.84)	4.800(121.92)
H	.965 (24.51)	.988 (25.10)
J	.690 (17.53)	.710 (18.03)
K	.415 (10.54)	.435 (11.05)
L	.140 (3.56)	.165 (4.19)
M	.020 (.51)	.030 (.76)
P	.314 (7.98)	.326 (8.28)
U	.025 (.64)	.046 (1.22)
V	.045 (1.14)	.070 (1.78)
Quality conformance inspection, part 3 (see note 7)		
A	3.335 (84.71)	3.365 (85.47)
B	.807 (20.50)	.817 (20.75)
D	2.250 (57.15)	2.300 (58.42)
E	2.195 (55.75)	2.380 (60.45)
F	3.410 (86.61)	3.550 (90.17)
N	.700 (17.78)	.800 (20.32)
R	55°	65°
S	115°	125°
T	.470 (11.94)	.530 (13.46)

NOTE:
Dimensions H, J, K, L, and M shall be maintained within .100 (2.54 mm) from the outer peripheral edge of the contact tabs.

Ⓑ FIGURE 1. Outline drawing of electron tube type 8168.

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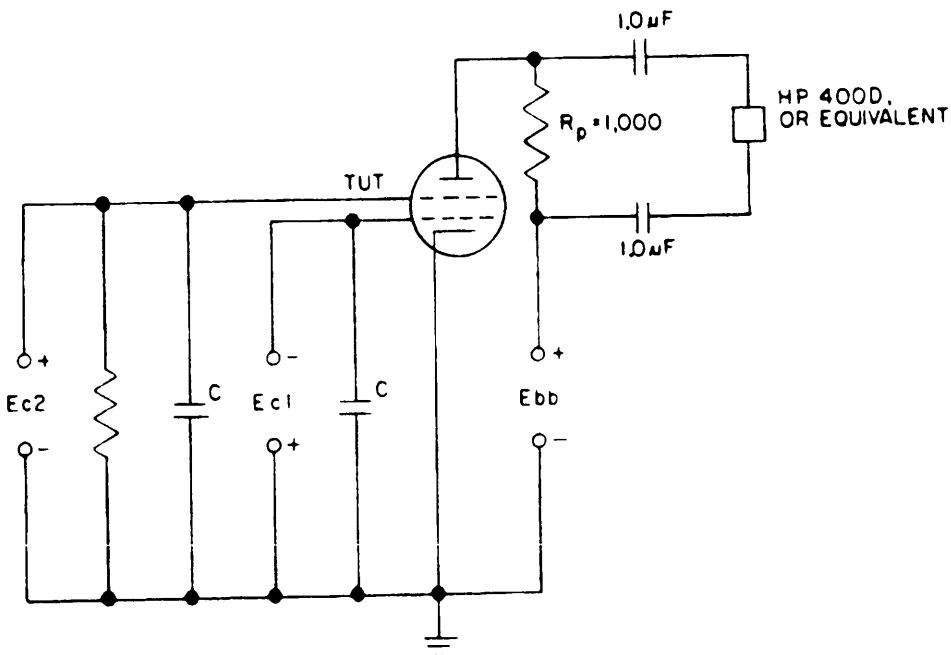


FIGURE 2. Basic circuit for vibration test (noise).

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